

Algorithms Design and Analysis (CSE-244)

Introduction to Algorithms



Dept. of Computer Science & Engineering

What is Algorithms?

- Definition: An algorithm is a sequence of unambiguous instructions for solving a problem.



Properties of an Algorithm

- Finite: the algorithm must eventually terminate
- Complete: Always give a solution when one exists
- Correct (sound): Always give a correct solution



Steps of an Algorithm Development Process

- Understand the problem
- Analyze the problem
- Plan a series of steps for solving the problem
- Review the algorithm/think about critical or corner cases
- Think about how those steps can be implemented/
choose the appropriate data structure



Classification of Algorithms

- Deterministic versus Randomized
- Exact versus Approximation
- Sequential/Serial versus Parallel



Some General Styles of Algorithms

- Brute Force (enumerative techniques, exhaustive search)
- Divide & Conquer
- Greedy Techniques
- Backtracking
- Dynamic Programming
- Graph Algorithms



Overview

Algorithms

Analysis

Designs Paradigm

Asymptotic
Analysis

Solving
Recurrence
Relations

Sorting
and
Searching

Divide &
Conquer

Greedy
Method

Backtrac
king

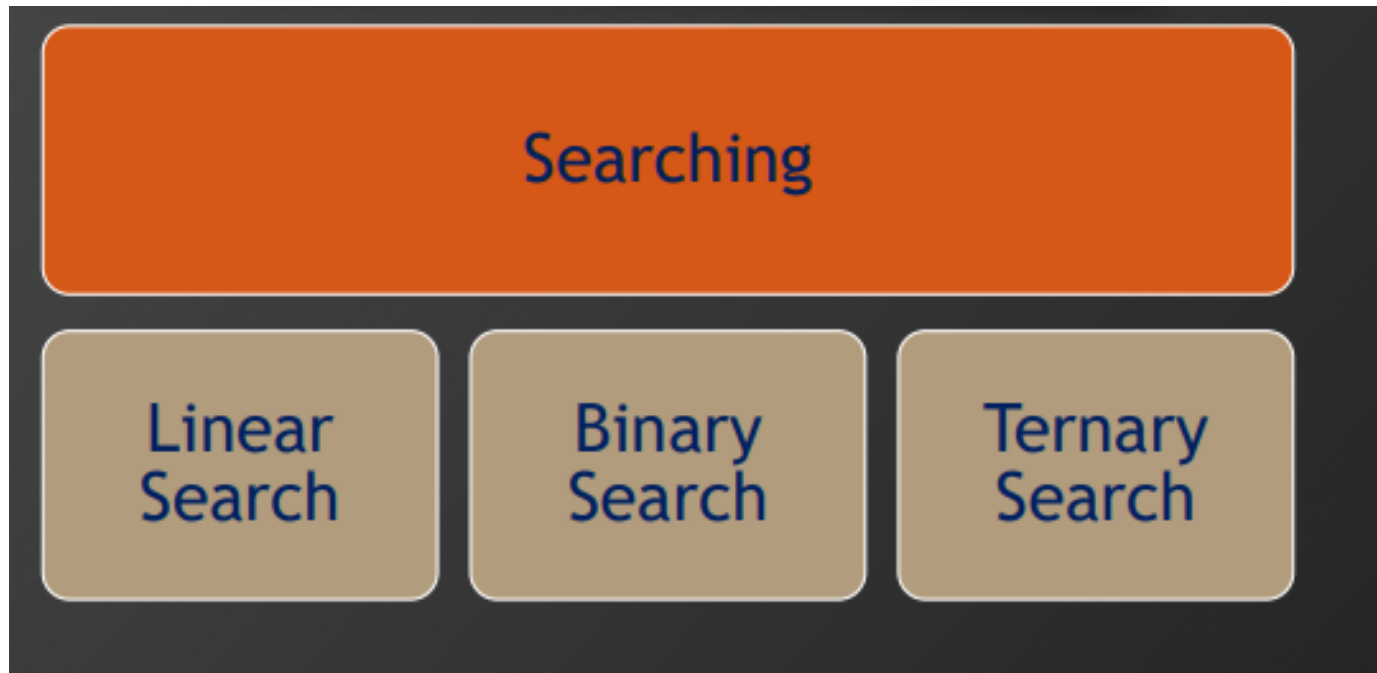
Dynamic
Program
ming

Sorting &
Searchin
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Graph
Algorith
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Searching Algorithms



Sorting Algorithms

Sorting Algorithms

Comparisons Sort

Non-Comparisons Sort

Bubble Sort

Quicksort

Heapsort

Shell sort

Mergesort

Introsort

Insertion sort

Selection sort

Odd-even sort

Cocktail shaker sort

Cycle sort

Merge insertion (Ford-Johnson) sort

Smoothsort

Timsort

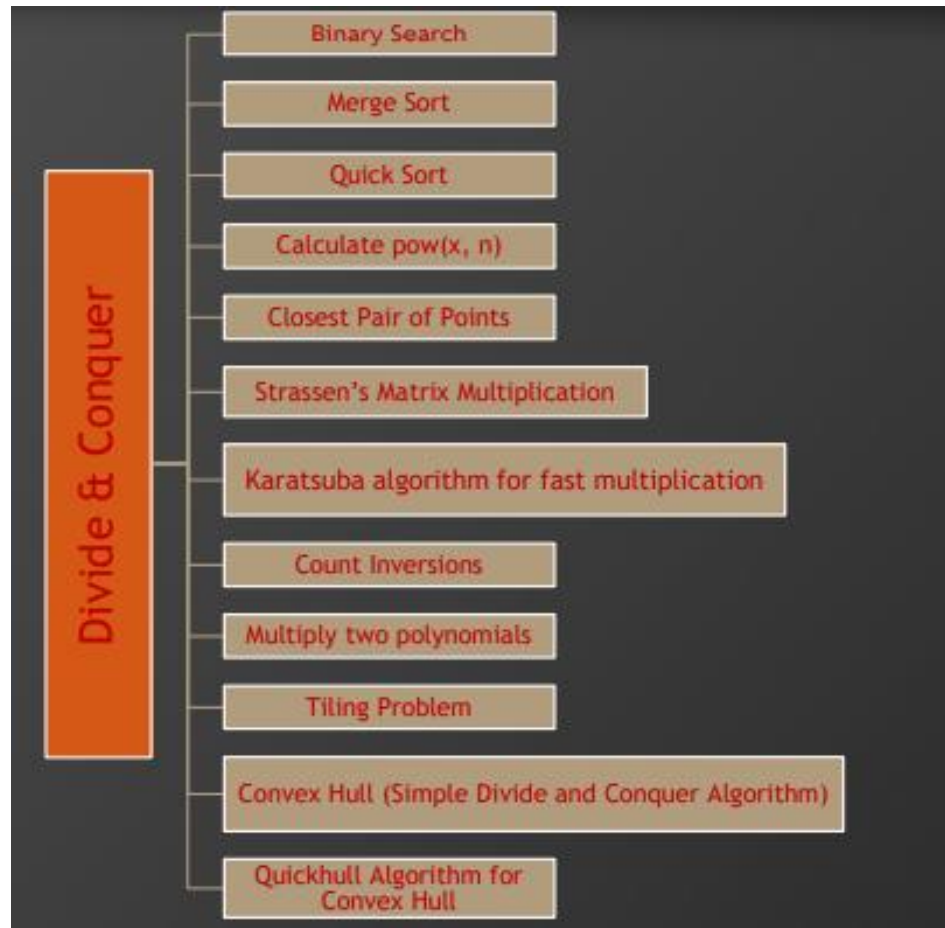
Counting sort

Radix sort

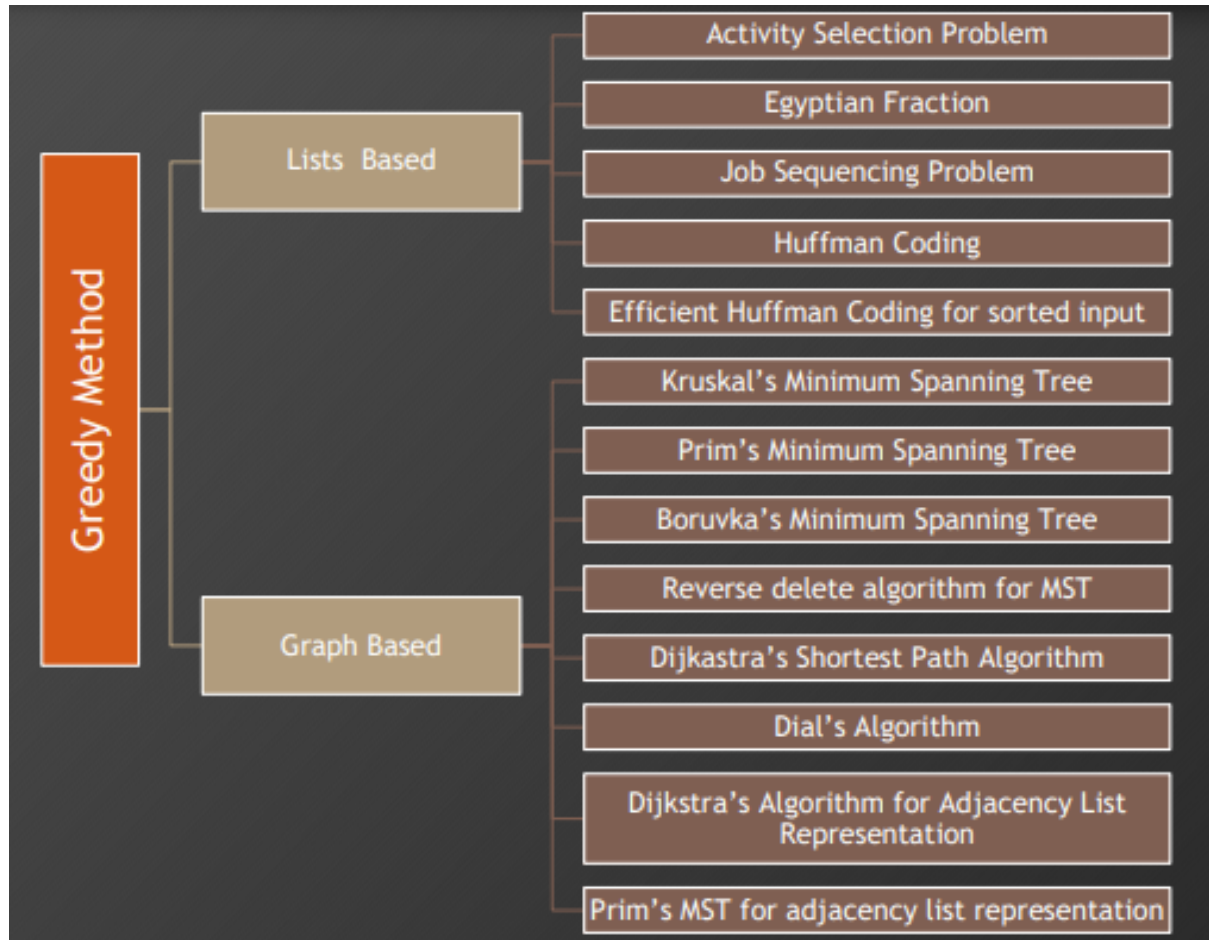
Bucket sort



Divide & Conquer



Greedy Methods



Backtracking



Dynamic Programming

Dynamic programming

Longest
Increasing
Subsequence

Longest
Common
Subsequence

Edit
Distance

Coin
Change

Matrix
Chain
Multiplication

0-1
Knapsack
Problem

Longest
Palindromic
Subsequence

Floyd
Warshall
Algorithm

Program
for
Fibonacci
numbers

Ugly
Numbers

Bellman-
Ford
Algorithm
for
Shortest
Paths

Largest
Independent
Set
Problem

Subset
Sum
Problem

Tiling
Problem



Graph Algorithms

Graph Algorithm

DFS and BFS

Minimum Spanning Tree

Breadth
First
Traversal
for a
Graph

Depth
First
Traversal
for a
Graph

Detecting
Cycle

Longest
Path

Topologic
al Sorting

Check
whether a
given
graph is
Bipartite
or not

Snake and
Ladder
Problem

Bi-
connecte
d Compone
nt

Prim's
Minimum
Spanning
Tree
(MST)

Applicatio
ns of
Minimum
Spanning
Tree
Problem

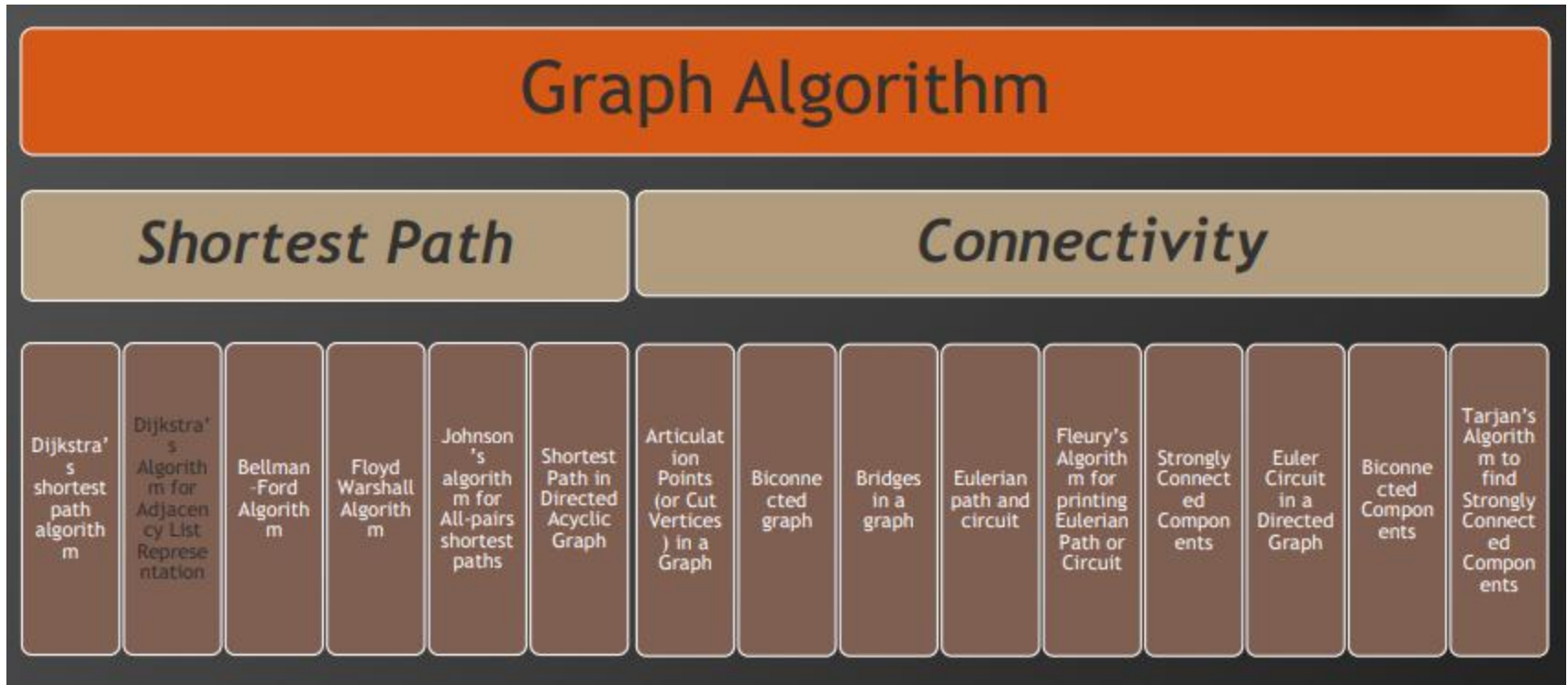
Prim's
MST for
Adjacency
List
Represent
ation

Kruskal's
Minimum
Spanning
Tree
Algorithm

Boruvka's
algorithm
for
Minimum
Spanning
Tree



Graph Algorithms



Graph Algorithms

Graph Algorithm

Maximum Flow

Ford-Fulkerson Algorithm for Maximum Flow Problem

Find maximum number of edge disjoint paths between two vertices

Find minimum s-t cut in a flow network

Maximum Bipartite Matching

Channel Assignment Problem

Hard Problems

Graph Coloring (Introduction and Applications)

Greedy Algorithm for Graph Coloring

Travelling Salesman Problem (Naive and Dynamic Programming)

Travelling Salesman Problem (Approximate using MST)

Hamiltonian Cycle

Vertex Cover Problem (Introduction and Approximate Algorithm)

K Centers Problem (Greedy Approximate Algorithm)



Happy Learning

